States of Matter

Exam code: 0620 & 0971

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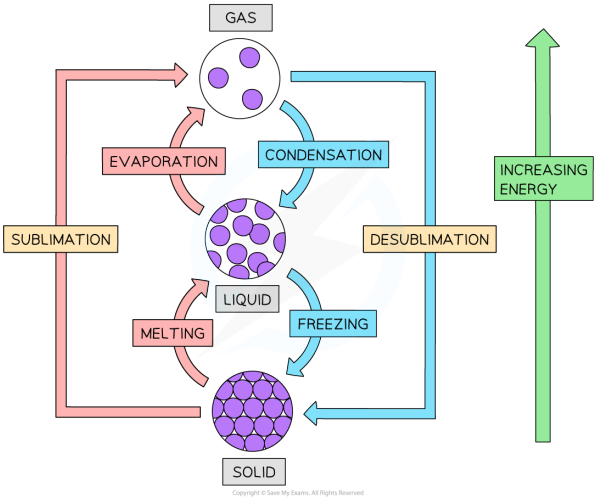
Help me understand

State changes

- **State changes** occur when:
  - Solids become liquids
  - Liquids become gases
  - Gases become liquids
  - Liquids become solids
- Each state change requires a change in the energy, arrangement and movement of the particles

The state changes



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Melting

- Melting is when a solid changes into a liquid
- Requires heat energy which transforms into **kinetic** energy, allowing the particles to move
- Occurs at a specific temperature known as the **melting point** (m.p.)

Freezing

- Freezing is when a liquid changes into a solid
- This is the reverse of melting and occurs at exactly the **same temperature** as melting, hence the melting point and freezing point of a pure substance are the same. Water, for example, freezes and melts at 0 °C
- Requires a significant decrease in temperature (or loss of thermal energy) and occurs at a specific temperature

Boiling

- Boiling is when a liquid changes into a gas
- Requires heat which causes bubbles of gas to form **below** the surface of a liquid, allowing for liquid particles to escape from the surface and within the liquid
- Occurs at a specific temperature known as the **boiling point** (b.p.)

Evaporation

- Evaporation occurs when a liquid changes into a gas and occurs over a **range** of temperatures
- Evaporation occurs only at the **surface** of liquids where high energy particles can escape from the liquid's surface at **low** temperatures, below the b.p. of the liquid
- The larger the surface area and the warmer the liquid surface, the more quickly a liquid can evaporate

Condensation

- Condensation occurs when a gas changes into a liquid on cooling and it takes place over a **range** of temperatures
- When a gas is cooled its particles lose energy and when they bump into each other they lack the energy to bounce away again, instead they group together to form a liquid

Examiner Tips and Tricks

Questions on the particle theory of matter show interconversion of states with a reversible arrow: ⇌, which means that the process can go forwards and backwards.

Read the question carefully and pick the direction of the change in state that the question refers to.

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State changes & kinetic theory

Extended tier only

- When substances are heated, the particles **absorb** thermal energy which is converted into kinetic energy
  - This is the basis of the kinetic theory of matter
- Heating a solid causes its particles to vibrate more
  - As the temperature increases, the particles vibrate so much that the solid **expands** until the structure breaks
  - This is when the solid **melts** into a liquid
- Heating a liquid causes its particles to move more and spread out
  - Some particles at the surface gain sufficient energy to overcome the intermolecular forces
  - This is when a liquid starts to **evaporate**
  - When the boiling point is reached, all of the particles gain enough energy to escape and the liquids **boils** into a gas
- These changes in state can be shown on a graph called a **heating curve**:

A heating curve showing the states, state changes and temperature changes as time progresses

- Cooling down a gas has the reverse effect and this would be called a **cooling curve**:

A cooling curve is like a heating curve, but is the mirror image

- Heating and cooling curves are used to show how changes in temperature affect changes of state
- The horizontal sections occur when there is a change of state but there is no change in temperature
- During a change of state, the thermal energy supplied is used to overcome the attractive forces between particles rather than increasing their kinetic energy
  - Since temperature is a measure of average kinetic energy, it remains constant until the state change is complete

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